

Uttkarsh Tyagi

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Data & cloud-focused engineer building streaming data pipelines, cloud-native ML systems, and production backend services. Recently completed **C-DAC's PG Certificate Programme in Big Data Analytics (PGCP-BDA)**. Comfortable across the stack: Python data/ML pipelines, AWS infrastructure via Terraform, and full-stack delivery.

EDUCATION

C-DAC — PG Certificate Programme in Big Data Analytics (PGCP-BDA)

Feb 2026 — Aug 2026

Uttarakhand Technical University, Dehradun

Sep 2021 — Jul 2025

B.Tech, Information Technology · CGPA 6.5

Delhi Public School, Ranipur, Haridwar

Class 12 — 88.4% · Class 10 — 86%

SKILLS

Languages	Python, TypeScript/JavaScript, SQL
Data Engineering	Apache Spark (Structured Streaming), Kafka/Redpanda, dbt, Delta Lake, medallion architecture (bronze/silver/gold), Apache Airflow
Cloud (AWS)	Lambda, S3, API Gateway, Step Functions, EventBridge, Kinesis Firehose, Glue Data Catalog, ECR, IAM, X-Ray, Terraform (IaC)
Machine Learning	scikit-learn, pandas, MLflow (tracking + registry), model evaluation & comparison pipelines, RAG pipelines (pgvector, SentenceTransformers), LLM API integration (Groq, Gemini, Ollama)
DevOps	Docker, Docker Compose, GitHub Actions (CI/CD), Terraform, Prometheus, Grafana
Databases	PostgreSQL, MongoDB, DynamoDB, schema design, indexing, query optimization
Tools	Git/GitHub, Postman, Linux, REST APIs, WebSockets

PROJECTS

liveflights — Real-time flight intelligence platform

github.com/tyxgx/liveflights · [live demo](#)

Spark · Redpanda · Delta Lake · dbt · Airflow · Terraform · AWS Lambda/S3/Step Functions · MLflow · FastAPI · Next.js

- Built an end-to-end streaming lakehouse — live flight data through Redpanda (Kafka-compatible) and Spark Structured Streaming into a medallion Delta Lake (bronze/silver/gold) on MinIO, modeled with dbt into Postgres marts.
- Designed and deployed a fully serverless replica of the same pipeline on AWS via Terraform: EventBridge-scheduled Lambda ingestion, Kinesis Firehose, S3 lake, Step Functions-orchestrated transforms, and a Glue Data Catalog — running live in production under a \$5/month budget.
- Orchestrated the pipeline with 4 Apache Airflow DAGs (hourly compaction, dbt, ML retraining, data-quality/drift checks), verified end-to-end (9/9 tasks green) in GitHub Actions CI; root-caused AWS account-level provisioning restrictions with IAM policy simulation and re-architected around them.
- Trained and served 4 ML models (corridor discovery, anomaly detection, traffic forecast, trajectory) on 28 days / 72M real ADS-B rows, validated on held-out days (1,831 corridors; forecast error down 41%), exposed live through a FastAPI + Next.js dashboard.

Global Music Intelligence Platform — Team Capstone (CDAC PGCP-BDA)

AWS Glue · Apache Spark · S3 · Athena · Terraform · GitHub Actions · PostgreSQL/pgvector · Django — 8-person team

- Independently built a parallel Bronze→Silver→Gold ETL pipeline in Terraform: Kaggle→EC2 ingestion, PySpark Glue jobs with incremental Silver loads, producing 5 Gold business tables — fully IaC, zero manual console steps, tested end-to-end.
- Independently designed and built the platform's AI-powered natural-language layer — the RAG chatbot and Django web application (StreamPulse, below) — end to end.

StreamPulse — Spotify analytics dashboard + RAG chatbot

github.com/tyxgx/steampulse

Django · DRF · PostgreSQL · pgvector · SentenceTransformers · Groq/Gemini/Ollama · AWS EC2 · Nginx — independently built consumption layer for the platform above

- Built an ETL pipeline (boto3 + pyarrow + psycopg2) loading Hive-partitioned Parquet from an S3 Gold data lake into Postgres, and the full RAG pipeline from scratch: chunked Gold rows, embedded ~215K chunks locally (SentenceTransformers, 384-dim), indexed with pgvector's ivfflat.
- Built 3 retrieval paths — keyword-routed SQL, per-entity scoped retrieval, and vector search with a confidence gate that declines rather than hallucinates — plus a 3-tier LLM fallback (Groq → Gemini → Ollama).
- Replaced the Glue/EC2 batch flow with a serverless DuckDB pipeline on GitHub Actions: incremental Bronze→Silver→Gold refresh to S3 (~42.8M source rows); rebuilt Gold matched the previous tables on 7,313/7,313 rows.
- Iteratively debugged retrieval quality against 15 fixed test questions, flipping 7/15 from wrong to correct with 0 regressions. Deployed end-to-end on AWS EC2 (Postgres 17 + pgvector, Unicorn + Nginx, systemd).

Interactive ML — Stage-by-stage ML pipeline explorer

github.com/tyxgx/interactive-ml · [live demo](#)

FastAPI · scikit-learn · Pyodide · pytest · Next.js · TypeScript · Tailwind CSS

- Built an educational ML platform exposing every pipeline stage — load, split, preprocess, train, predict, evaluate — as an inspectable step, with leakage-safe preprocessing and a fair 5-algorithm comparison evaluated on an identical train/test split per session.
- Moved the Python backend into the browser (Pyodide Web Worker running the same FastAPI code); added model diagnostics (learning curve, ROC/PR, permutation importance), 38 pytest tests and GitHub Actions CI.

EXPERIENCE

Full Stack Developer Intern — Unlock Discounts, Bangalore (Remote)

Jul 2025 — Oct 2025

- Built a customer review section (React/JS/CSS) and APIs to fetch and display data from the company's database across platforms; created a certificate verification system for intern authentication.

Project Intern — BHEL, Haridwar

Jul 2025 — Aug 2025

- Developed a web portal (HTML/CSS/React, SQL backend) for equipment issuance with late tracking and recipient records.